

More Serious Algebra I
 26/27 Semester 1
 Compiled at 09:40, Friday 9th October, 2026

Week	Concepts (Not necessarily accurate)
3	Group axioms, subgroup tests, cyclic groups, element order
4	Dihedral group D_n , products, inverses, cyclic subgroups, orders
5	Subgroup order, generators, cyclic groups
6	Left/right cosets, partitions, index, Lagrange's theorem
7	Normality tests, cosets, quotient operations
8	Kernel, image, injectivity/surjectivity, normal subgroups, quotient groups
9	Everything from Weeks 3–9
10	Ring axioms, units, zero divisors, subring tests
11	Ideals, principal ideals in familiar rings, subrings vs. ideals
12	Kernels, images, quotient rings
13	Divisibility, degree, remainders, roots/factors
14	GCD, Bézout identity, polynomial division

Format.

- (1) Four problems each week for two categories: Starter and Advanced.
- (2) Four easy for Starter.
- (3) One low-medium, two medium, and one hard for Advanced.
- (4) The problems will be given on the spot.

Time and Location. Tuesday, 2:00-3:40 PM, any empty room.

Link.

Clickable QR code to link

Unused Problems

This is the current draft of the list of rejected candidates.

1 (E). Let G be a group. Show that $(ab)^{-1} = b^{-1}a^{-1}$ for all $a, b \in G$.

2 (L-M). A group that has only a finite number of subgroups must be finite.

3 (M). Let G be a nonempty finite set with an associative binary operation such that for all $a, b, c \in G$, $ab = ac \Rightarrow b = c$ and $ba = ca \Rightarrow b = c$ (in other words, G satisfies left and right cancellation laws). Then G is a group. Show that this conclusion may be false if G is infinite.

4 (H). Let the group G be the union of finitely many, let us say n , cosets of (not necessarily distinct) subgroups $C_1, C_2, \dots, C_n$:

$$G = \bigcup_{i=1}^n C_i g_i.$$

Then at least one subgroup C_i has finite index in G .

5 (L-M). Assume $G = \{1, a, b, c\}$ is a group of order 4 with identity 1. Assume also that G has no elements of order 4. Use the cancellation laws to show that there is a unique Cayley table for G . Deduce that G is abelian.

6 (E). Let G be a finite group.

(a) Let $a \in G$. Show that there is a positive integer n such that $a^n = e$, the identity element.

(b) Show that there exists a positive integer n such that $a^n = e$ for all $a \in G$.

7 (H). Let G be the set of integers modulo 2^n with operation

$$x * y = 4xy + x(-1)^y + y(-1)^x \pmod{2^n}.$$

Show that G is a cyclic group.